



**FACULTY OF INFORMATION TECHNOLOGY
BACHELOR OF SCIENCE IN INFORMATICS
ICS1203: OBJECT ORIENTED PROGRAMMING I**

END OF SEMESTER EXAMINATION

DATE: 7th November 2016

TIME: 2 HOURS

INSTRUCTIONS Attempt Question ONE and any other TWO questions

Question One

- a) Using examples, differentiate between the following programming paradigms and object oriented programming. [4 Marks]
- i) Unstructured programming
 - ii) Modular Programming
- b) Using simple java code segments, briefly explain the following object oriented programming concepts.
- i) Constructors
 - ii) Polymorphism
 - iii) Encapsulation
 - iv) Delegation [8 Marks]
- c) Using simple java code segments differentiate between the following as used in object oriented programming.
- i) Class variable and instance variable
 - ii) Parameter passing by value and by reference
 - iii) Method over-loading and Method over-riding
 - iv) Interface and abstract Class [8 Marks]

Question Two

- a) Using relevant examples in java, explain the following inherited members of a class:
- i) Public members of the super class.
 - ii) Protected Members of the super class.
 - iii) Constructors of the super class. [3 Marks]
- b) You have been approached by the Human Resource Department to develop their system to keep track of employee's payments. Each employee has the following :
- Details: Name, Personal File Number (automatically generated based on the number of employees since inception), Gross Salary.
 - The operations: compute net payments to an employee after tax. Tax is computed as 30% of the gross salary.

You are required to write Java code to create:

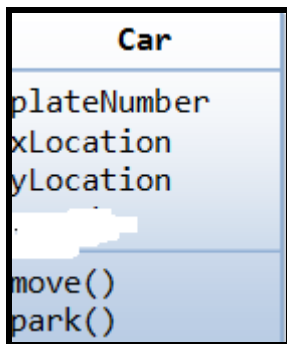
- ii) A class **Employee**, for the above with relevant methods to retrieve and change their values. [7 Marks]
- iii) Define two new classes of special types of employees:
 - Full-time employees who are paid a fixed salary and can take an advance (more than their salary).
 - Part-time employees whose salary is based on the number of hours worked and cannot be paid in advance (more than their dues). [6 Marks]
- iv) A main method to instantiate the above and show accordingly [4 Marks]

Question Three

- a) Using simple java code segments, briefly explain the concept of Exception Handling. [4 Marks]
- b) Suppose you are to design a system for Loan for customers in a BANK. Once the customer applies for a loan, an account is created for him/her with the loan amount approved and the interest rate to be charged. Each month the customer makes repayment of the loan and a new balance is calculated based on the new principal amount remaining and the interest charged on it. Create a class called Loan with: customer name, Customer Number, the loan balance and an operation to compute the new balance after repayment. [8 Marks]
- c) Create a Graphical User Interface (GUI) to input and manipulate the details of Objects of Loan Class above. [8 Marks]

Question Four

A program is required to track a Car. A car has the properties shown below.



XLocation and yLocation is the distance on x-axis and y-axis respectively from the initial position at flag off point (`xLocation = xLocation_initial`, `yLocation = yLocation_initial`). The Car can change its position by moving the x and y coordinates or park (return to flag off point). You are required write java code segments to create the following:

- a) A Class called **Car** that has the member above. Include methods to retrieve the location of the car (xlocation and ylocation) and the distance of the car from the flag off point. [8 Marks]
- b) A method in class **Car** to compute the distance from a car c1 from another car c2 on a straight road. [4 Marks]
- c) Overload the method b) above such that the road is semi-circular. [3 Marks]
- d) A main method to instantiate the class objects and relevant method calls [5 Marks]

Question Five

Suppose that you wish to develop a client system with the data as shown below.

Orders				
Customer ID	Customer Name	Unit Price	Quantity	Total
23367	Otieno Ltd	77.80	30	
23456	Wanjiru Orare	6.50	400	
23681	Strathmore St	98.00	700	
21456	Wakulima Wote	42.50	56000	

- a) Briefly explain how to create the application and connect to a database. [4 Marks]
- b) Write Java code segments to achieve the following:
- i) Create a User interface to input the above data and compute the total value based on the Unit Price and Quantity. [8 Marks]
- ii) Store the data in a database and be able to retrieve it. [8 Marks]